Solution to Exercise Sheet 5:
Empirical distributions and learning

Foundations of Neuro-Symbolic AI
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Task 1:
The empirical distribution $\mathbb{P}^D[X_0, X_1, X_2]$ is the average of the one-hot encodings of each observed state tuple.

(i) We introduce a decomposition variable I taking states in [10] enumerating the days. Based on this variable we find a CP decompositon

$$\mathbb{P}^D[X_0, X_1, X_2] = \left\langle \mathbb{P}\left[X_0|I\right], \mathbb{P}\left[X_1|I\right], \mathbb{P}\left[X_1|I\right], \mathbb{P}[I] \right\rangle_{[X_0,X_1,X_2]}$$

where $\mathbb{P}[I] = \frac{1}{10}\mathbb{I}[I]$ and the leg matrices are

$$\mathbb{P}\left[X_0|I\right] = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} [X_0, I]$$
$$\mathbb{P}\left[X_1|I\right] = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix} [X_1, I]$$
$$\mathbb{P}\left[X_2|I\right] = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \end{pmatrix} [X_1, I] \, .$$

(ii) We want to find the contraction of $\mathbb{P}^D[X_0, X_1, X_2]$ with the propositional formula

$$f[X_1, X_2] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} [X_1, X_2] \, ,$$

representing the event that either both Alice and Bob bring and umbrella or both do not bring an umbrella. The rate of days on whis this is true is the contraction

$$\left\langle \mathbb{P}^D[X_0, X_1, X_2], f[X_1, X_2] \right\rangle_{[\varnothing]}$$

We use the CP decomposition above and can ignore the core $\mathbb{P}\left[X_0|I\right]$ since X_0 (Rain) does not appear. Further, we notice, that $\mathbb{P}[I]$ only brings a factor of $\frac{1}{10}$. We are left with computing

$$\frac{1}{10} \cdot \left\langle \mathbb{P}\left[X_1|I\right], \mathbb{P}\left[X_2|I\right], f[X_1, X_2] \right\rangle_{[\varnothing]} = \frac{5}{10} = \frac{1}{2} \, .$$

Task 2:
The empiricial distribution is the order 3 tensor

$$\mathbb{P}^D[X_0, X_1, X_2] = \frac{1}{10} \cdot \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} [X_0, X_1, X_2] \, .$$

- We notice that

$$\mathbb{P}^D\left[X_0, X_1, X_2 = 0\right] = \mathbb{P}^D\left[X_0, X_1, X_2 = 1\right]$$

  and therefore

$$\mathbb{P}^D\left[X_0, X_1, X_2\right] = \mathbb{P}^D\left[X_0, X_1\right] \otimes \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} \left[X_2\right]$$

  which means $(X_2 \perp X_0, X_1)$.

- We are left to check $(X_1 \perp X_0)$. For this we compute the marginals

$$\mathbb{P}^D\left[X_0, X_1\right] = \frac{1}{10} \cdot \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix} \left[X_0, X_1\right]$$

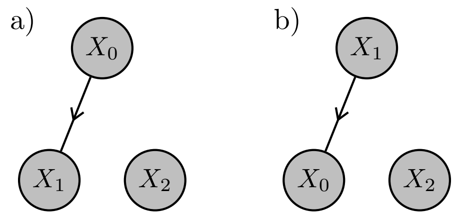
  and

$$\mathbb{P}^D\left[X_0\right] = \frac{1}{10} \cdot \begin{pmatrix} 4 & 6 \end{pmatrix} \left[X_0\right] \quad , \quad \mathbb{P}^D\left[X_1\right] = \frac{1}{10} \cdot \begin{pmatrix} 4 & 6 \end{pmatrix} \left[X_0\right] .$$

  We have $\mathbb{P}^D\left[X_0, X_1\right] \neq \mathbb{P}^D\left[X_0\right] \otimes \mathbb{P}^D\left[X_1\right]$ and thus $(X_1 \perp X_0)$ does not hold.

Task 3:
Notice, that in the unregularized learning setup, the solution is the empirical distribution itself represented as a Bayesian network.

(i) There are two directed hypergraphs capturing the independence statement $(X_2 \perp X_0, X_1)$



(ii) Based on the calculations in task 2 the optimal decorating conditional distributions are in the case $a)$

$$\mathbb{P}^D\left[X_1 | X_0\right] = \begin{pmatrix} \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{2}{3} \end{pmatrix} \left[X_1, X_0\right]$$
$$\mathbb{P}^D\left[X_2\right] = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} \left[X_2\right]$$
$$\mathbb{P}^D\left[X_0\right] = \begin{pmatrix} \frac{2}{5} \\ \frac{3}{5} \end{pmatrix} \left[X_0\right]$$

In case $b)$ the optimal decorating conditional distributions would be

$$\mathbb{P}^D\left[X_0 | X_1\right] = \begin{pmatrix} \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{2}{3} \end{pmatrix} \left[X_0, X_1\right]$$
$$\mathbb{P}^D\left[X_2\right] = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} \left[X_2\right]$$
$$\mathbb{P}^D\left[X_1\right] = \begin{pmatrix} \frac{2}{5} \\ \frac{3}{5} \end{pmatrix} \left[X_1\right]$$

Task 4:
With respect to the extended dataset the empirical distribution is then the tensor

$$\mathbb{P}^D\left[X_0, X_1, X_2\right] = \tfrac{1}{11}\cdot\begin{pmatrix}1 & 1\\ 1 & 2\end{pmatrix}\begin{pmatrix}1 & 1\\ 1 & 3\end{pmatrix}\left[X_0, X_1, X_2\right] .$$

The independence statement $(X_2 \perp X_0, X_1)$ does not hold any more, since

$$\mathbb{P}^D\left[X_0 = 1, X_1 = 1, X_2 = 1\right] = \frac{3}{11} \neq \frac{5}{11}\cdot\frac{6}{11} = \mathbb{P}^D\left[X_0 = 1, X_1 = 1\right]\cdot\mathbb{P}^D\left[X_2 = 1\right] .$$

Also no conditional independence statement holds:

(i) $(X_0 \perp X_1) \mid X_2$ does not hold since

$$\mathbb{P}^D\left[X_0, X_1 | X_2 = 1\right] = \frac{1}{6}\cdot\begin{pmatrix}1 & 1\\ 1 & 3\end{pmatrix}\left[X_0, X_1\right] \neq \frac{1}{6}\cdot\begin{pmatrix}2\\ 4\end{pmatrix}\left[X_0\right] \otimes \frac{1}{6}\cdot\begin{pmatrix}2\\ 4\end{pmatrix}\left[X_0\right] .$$

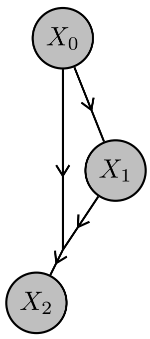
(ii) $(X_1 \perp X_2) \mid X_0$ does not hold since

$$\mathbb{P}^D\left[X_1, X_2 | X_0 = 1\right] = \frac{1}{7}\cdot\begin{pmatrix}1 & 2\\ 1 & 3\end{pmatrix}\left[X_1, X_2\right] \neq \frac{1}{7}\cdot\begin{pmatrix}3 & 4\end{pmatrix}\left[X_1\right] \otimes \frac{1}{7}\cdot\begin{pmatrix}2 & 5\end{pmatrix}\left[X_2\right] .$$

(iii) $(X_0 \perp X_2) \mid X_1$ does not hold since

$$\mathbb{P}^D\left[X_0, X_2 | X_1 = 1\right] = \frac{1}{7}\cdot\begin{pmatrix}1 & 1\\ 2 & 3\end{pmatrix}\left[X_0, X_2\right] \neq \frac{1}{7}\cdot\begin{pmatrix}2 & 5\end{pmatrix}\left[X_0\right] \otimes \frac{1}{7}\cdot\begin{pmatrix}3 & 4\end{pmatrix}\left[X_2\right] .$$

The Bayesian networks in task 3 would be the naive chain decompositions (see lecture II.1-Independence) of the empirical distributions, with respect to one of the 6 possibilities to order the three variables. When choosing the order X_0, X_1, X_2 this is the hypergraph



which would be decorated by the conditional probability distributions

$$\mathbb{P}^D\left[X_0\right] = \frac{1}{11}\cdot\begin{pmatrix}4\\ 7\end{pmatrix}\left[X_0\right] \quad , \quad \mathbb{P}^D\left[X_1 | X_0\right] = \begin{pmatrix}\frac{1}{2} & \frac{1}{2}\\ \frac{2}{7} & \frac{5}{7}\end{pmatrix}\left[X_1, X_0\right]$$

and

$$\mathbb{P}^D\left[X_2 | X_0, X_1\right] = \begin{pmatrix}\frac{1}{2} & \frac{1}{2}\\ \frac{1}{2} & \frac{2}{5}\end{pmatrix}\begin{pmatrix}\frac{1}{2} & \frac{1}{2}\\ \frac{1}{2} & \frac{3}{5}\end{pmatrix}\left[X_0, X_1, X_2\right] .$$